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## Task 3: Matrix Operations Using NumPy

### 1. Introduction

Matrix operations are an important part of numerical computing and scientific applications. Python provides the NumPy library, which supports efficient matrix operations using arrays. In this task, basic matrix operations are performed using NumPy.

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### 2. Objective

The objectives of this task are:

- To understand matrix representation using NumPy
  - To perform basic matrix operations using Python
  - To implement mathematical concepts programmatically
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### 3. Tools and Technologies Used

- **Programming Language:** Python
  - **Library Used:** NumPy
  - **IDE:** Visual Studio Code
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### 4. Methodology

1. Two matrices were created using NumPy arrays.
  2. Matrix addition and subtraction were performed using arithmetic operators.
  3. Matrix multiplication was performed using the `dot()` function.
  4. Transpose of the matrix was obtained using the transpose property.
  5. Determinant of the matrix was calculated using the `linalg.det()` function.
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### 5. Results and Observations

- Matrix addition and subtraction were performed successfully.
- Matrix multiplication produced a valid resultant matrix.
- The transpose of the matrix interchanged rows and columns.

- The determinant value was calculated correctly using NumPy.
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## **6. Conclusion**

This task demonstrated the implementation of basic matrix operations using NumPy. It helped in understanding how mathematical concepts such as matrix multiplication, transpose, and determinant can be implemented using Python. NumPy simplifies matrix calculations and improves computational efficiency.

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## **7. Output**

- Python program (matrix\_operations.py)
- Terminal output showing matrix operation results